

Data Collection Report

Project: Mobile Money Transaction Pattern Analysis

Target: User Activity Classification (Low / Medium / High)

1. Sampling Strategy

We employed **purposive stratified sampling** to ensure diversity across key dimensions: age groups (18–65), gender, profession, geographic zone (urban/suburban/rural), and income level. Each of the 5 group members was responsible for recruiting at least 10 participants from different backgrounds, resulting in a total sample of 60 participants — exceeding the required minimum of 50.

Participants were recruited through personal networks, university campuses, local markets, community centers, and professional contacts. Snowball sampling was used where direct access was limited.

2. Data Sources and Variables

Primary data was collected through a structured questionnaire administered face-to-face and via WhatsApp voice/video calls. Transaction history summaries were self-reported by participants referencing their mobile money transaction histories (last 3–12 months). No actual account numbers, PINs, or exact balances were recorded.

Key variables collected: age, gender, profession, education level, household size, monthly income range, geographic zone, smartphone ownership, mobile money provider, years of usage, primary use (personal/business), monthly transaction counts, transaction types, and approximate amounts.

3. Ethical Considerations

- All participants signed an informed consent form before data collection began.
- Participants were assigned unique anonymous IDs (USER_001 to USER_060). No names or phone numbers were recorded.
- Participation was entirely voluntary; participants could withdraw at any time.
- Data is stored in password-protected files, accessible only to group members.
- The study posed no risk of financial, social, or psychological harm to participants.

4. Data Quality Assurance

To ensure data quality, each questionnaire was reviewed immediately after collection for completeness. Ambiguous or inconsistent responses were clarified with participants on the spot where possible. After digitization, a second team member independently verified a 20% random sample of entries.

Known limitations: Transaction amounts are self-reported estimates and may not perfectly reflect actual records. Recall bias is possible for transactions more than 3 months old. The sample may slightly overrepresent urban and educated users due to recruitment methods.

5. Collection Challenges

- Some participants were reluctant to share transaction-related data; trust was built by emphasizing anonymization.
- Rural participants required field visits, increasing time and logistics costs.
- Estimating monthly transaction amounts from memory introduced approximation errors, mitigated by asking participants to review their phone's transaction history.
- Some older participants needed assistance navigating their phone's transaction history.